

Curriculum Vitae - Evan Toon - Undergraduate Physics Student

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EDUCATION

University of Tennessee, Expected Graduation May 2027

Knoxville, TN

Physics(BS), Academic Concentration, Astronomy Minor

- GPA: Undergraduate - 3.89

RESEARCH EXPERIENCE

Dynamic Nuclear Polarization (DNP) - Nuclear Physics Research

April 2025 - Present

Dr. Nadia Fomin - University of Tennessee

Knoxville, TN

- **Computational** - Worked closely with Professor Fomin and collaborators from Jefferson Lab and Oak Ridge National Lab to further investigate how the nuclear polarization of a sample is measured via a Nuclear Magnetic Resonance (NMR) coil within a DNP characterization apparatus by performing a three-Gaussian fitting analysis to infer the physical meaning of a misunderstood background signal that contributes to the overall polarization measurement.
- **In the Lab** - Worked with Professor Fomin and her graduate students to recommission a DNP target setup transferred from Oak Ridge National Lab. Actively contributed to the reassembly, optimization, and operation of the system, gaining hands on expertise with the dilution refrigerator, superconducting magnet, vacuum systems, and other critical DNP components.

Science Undergraduate Laboratory Internships (SULI) - DNP Research at HFIR) May 2026 - August 2026

Dr. Josh Pierce - Oak Ridge National Laboratory (ORNL)

Oak Ridge, TN

- Actively contributed to the commissioning and optimization of a new Dynamic Nuclear Polarization (DNP) system for the IMAGINE-X BRAVE protein crystallography project at the High Flux Isotope Reactor (HFIR). I worked on the design and construction of a microwave equipped sample stick with integrated NMR circuitry, performed polarization measurements and optimization studies on multiple samples of interest, and gained hands on experience operating a 1 K ^4He evaporation refrigerator, 5 T superconducting magnet, microwave systems, and NMR instrumentation.

PRESENTATIONS

Selected Talks

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| (1) APS Division of Nuclear Physics Fall Meeting - Undergraduate Oral Presentation | Fall 2025 |
| (2) UTK Department of Physics & Astronomy Summer Research Fellowship Summary | Fall 2025 |
| (3) SPS Zone 8 Meeting - Oral Presentation | Spring 2026 |
| (4) UTK Arts & Sciences Undergraduate Research Symposium (ASURoS) | Spring 2026 |

Selected Posters

- | | |
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| (1) UTK Exhibition of Undergraduate Research and Creative Achievement (EURC) | Spring 2026 |
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RELEVANT COURSES

- | | |
|---|------------------------|
| Classical Mechanics I [PHYS 311]; Classical Mechanics II [PHYS 312] | Fall 2025, Spring 2026 |
| Waves, Optics, & Relativity [PHYS 251]; Quantum Physics [PHYS 252] | Spring 2025, Fall 2025 |
| Astrophysics I [ASTR 217]; Astrophysics II [ASTR 218] | Fall 2024, Spring 2025 |
| Multivariable Calculus [PHYS 201]; Linear Algebra [PHYS 301] | Fall 2025, Spring 2026 |

TECHNICAL PROFICIENCIES

Programming: Python (NumPy, Matplotlib, Pandas, SciPy, lmfit), Git

Software: L^AT_EX, Mathematica

AWARDS & HONORS

Dean's List

Fall 2023 – Spring 2026

COMMUNITY & LEADERSHIP

Society of Physics Students (SPS)

- *Outreach/Media Coordinator*

- *President*

Fall 2023 – Present

Spring 2026

Fall 2026 – Spring 2027

Astrophysics Journal Club

- *Member*

Fall 2023 – Present

Astronomy Club

- *Member*

Fall 2025 - Present

Phi Beta Kappa (ΦBK)

- *Member*

Spring 2026 - Present

Downhill Longboarding Event Organizer

- *Planned, organized, and hosted several successful downhill longboarding race/freeride events*

2021 - Present

INTERESTS

Experimental Nuclear Physics

Dynamic Nuclear Polarization (DNP)

Cryogenic Instrumentation

- *Dilution Refrigeration, 4He Evaporation Refrigerators, etc*

Solid Polarized Scattering Targets

Quantum Chromodynamics (QCD), Probing Nucleon Structure

Electron Scattering, Neutron Scattering

Low Temperature Physics

Neutron Crystallography

Experimental Cosmology